

Interview Q&A: numpy

1. What is NumPy?

NumPy is a library for working with arrays and mathematical operations.

2. What are the main benefits of using NumPy arrays over Python lists?

NumPy arrays provide faster computation, better memory management, and more efficient data operations compared to Python lists.

3. What is the difference between `numpy.array()` and `numpy.asarray()`?

`numpy.array()` creates a new array, while `numpy.asarray()` attempts to convert an existing object into an array.

4. What is the difference between numpy's random number generation functions, `np.random.rand()` and `np.random.randn()`?

`np.random.rand()` generates random numbers from a uniform distribution, while `np.random.randn()` generates random numbers from a standard normal distribution.

5. What is the purpose of numpy's NaN value?

NaN stands for Not a Number, it's used to represent undefined or unreliable results in numerical computations.

6. What are some common use cases for NumPy arrays?

NumPy arrays are commonly used for numerical computations, data analysis, scientific computing, and machine learning.

7. How do you create an empty numpy array?

You can use `numpy.empty()` function to create an empty numpy array.

8. How do you get the shape of a NumPy array?

You can use the `.shape` attribute.

9. How do you create a 2D NumPy array from a Python list?

You can use the `numpy.array()` function with the input list, specifying the desired data type.

10. What is the difference between numpy's broadcasting rules for 1D arrays versus 2D arrays?

In general, NumPy's broadcasting rules are more flexible for 2D arrays than 1D arrays.

11. What is the purpose of numpy's zeros() function?

The `numpy.zeros()` function creates an array filled with zeros.

12. How do you create a NumPy array from a Python list?

You can use the `numpy.array()` function to convert a Python list into a NumPy array.

13. What is the difference between numpy's vectorized operations and Python's list comprehensions?

Vectorized operations are faster and more memory-efficient, but can be less readable.

14. How do you create a NumPy array from a Python list of integers?

You can use the `numpy.array()` function to convert a Python list into a NumPy array.

15. How do you create a NumPy array from a Python list of floats?

You can use the `numpy.array()` function to convert a Python list of floats into a NumPy array.

16. What is the difference between numpy's dtype objects, such as int32 and float64?

numpy's dtype objects define the data type of a NumPy array, including its size and precision.

17. What is the difference between numpy's mean() function and np.mean()?

`np.mean()` is a function that calculates the arithmetic mean of an array, while `mean()` is a method that can be applied to a specific axis of a multi-dimensional array.